FOCUS: Object-Centric Evidence as a Causal Update Gate for Realistic Online Vision-Language Adaptation

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Abstract

Realistic online test-time adaptation for CLIP-style vision-language models is fragile because each accepted pseudo-labeled sample can permanently alter the adaptation state. We revisit a narrow hypothesis: can lightweight object-centric evidence act as a safer write signal for STATA-style online adaptation? FOCUS builds a full-image view, a foreground view, and a complementary background view, then gates STATA updates using agreement and foreground-versus-background support. Following reviewer feedback, we make validation-derived causal thresholds the primary protocol and relegate exact-rate top-K selection to an offline oracle-style ranking analysis. On the causal protocol at the common nominal 50% rate, FOCUS is the strongest causal gate on WATERBIRDS, reaching 71.03% online accuracy and 46.24% worst-group accuracy, versus 62.04%, 62.04%, and 63.28% for entropy, max-probability, and CER gating. The same method fails on COUNTERANIMAL, where it reaches 62.30% and trails entropy (69.44%), max-probability (67.93%), and CER (66.75%). The mechanistic picture is also negative: FOCUS harmful-update AUPRC remains 0.0227 on WATERBIRDS and 0.0089 on COUNTERANIMAL, below stronger lightweight baselines, and a background-only ablation outperforms the full score downstream. Correcting the protocol and latency accounting therefore changes the paper's conclusion: object-centric evidence is an intuitively plausible but insufficient standalone update-safety signal for realistic online VLM adaptation.

1 Introduction

Online test-time adaptation (TTA) for vision-language models (VLMs) is appealing because it promises stream-time adjustment without retraining. In realistic online settings, however, the central difficulty is not merely making a better prediction on the current image; it is deciding whether the current pseudo-labeled sample is safe to write into the running state at all. Zanella et al. [2025] show that this decision matters under temporally correlated streams, while Dobler et al. [2024] and Sheng et al. [2025] show how easily headline TTA gains collapse under stricter evaluation.

This paper studies one deliberately narrow hypothesis. We ask whether *object-centric evidence* can improve the write decision for STATA [Zanella et al., 2025]. The intuition is simple: if the predicted class is supported by a foreground view but not by the complementary background, the sample may be safer for online adaptation than one whose evidence is mostly contextual.

The original version of this study made its matched-rate comparisons by sorting scores over the full test stream and accepting the top-K samples. That protocol is useful as an offline ranking diagnostic, but it is not causal: it uses future information unavailable to a real online system. In this revision, the main results therefore use *validation-derived thresholds* only. Exact-rate runs are retained, but explicitly as offline oracle-style ranking analysis.

Under the corrected protocol, the story is mixed and ultimately negative. On WATERBIRDS, FOCUS is the strongest causal gate by downstream accuracy: at the nominal 50% rate it reaches 71.03% accuracy and 46.24% worst-group accuracy, versus 62.04%/31.07% for entropy, 62.04%/31.07%

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for max-probability, and 63.28%/31.66% for CER. But the mechanistic target is not met: FOCUS harmful-update AUPRC remains 0.0227, below max-probability (0.1026) and CER (0.1935). On COUNTERANIMAL, FOCUS reaches 62.30% overall accuracy and 61.29% balanced accuracy, while entropy, max-probability, and CER reach 69.44%/69.05%, 67.93%/68.00%, and 66.75%/66.45%, respectively. A background-only ablation further outperforms the full score downstream on both datasets.

Our contributions are:

- We reformulate FOCUS as a causal online update-gating study for STATA, with validation-derived thresholds as the primary evaluation and exact-rate top-K selection explicitly demoted to offline ranking analysis.
- We report matched-threshold experiments on WATERBIRDS and COUNTERANIMAL with sparse counterfactual harmful-update labels, simple object-centric ablations, and corrected latency accounting that separates cached gate latency from full end-to-end cost.
- We show that the central hypothesis is not supported overall: object-centric evidence can help on WATERBIRDS, but it does not yield the best harmful-update ranking, it fails on COUNTERANIMAL, and its apparent gains are partly reproduced by simpler background-only filtering.

Section 2 reviews related work, Section 3 defines FOCUS and the corrected protocols, Section 4 presents results, Section 5 discusses implications and limitations, and Section 6 concludes.

2 Related Work

Realistic online VLM adaptation. Our setting follows Zanella et al. [2025], which argues that VLM TTA should be evaluated under temporally correlated online streams and introduces STATA as a strong statistics-based online baseline. Dobler et al. [2024] and Sheng et al. [2025] both caution that many VLM TTA gains weaken under stronger comparisons. Those papers motivate the narrow scope of this work: we study update selection for a fixed adapter rather than claiming a new TTA family.

Reliability-oriented update filtering. Recent VLM TTA work increasingly focuses on which pseudo-labels should influence the model. ReTA uses consistency-aware reliability signals [Liang et al., 2025]; other work studies uncertainty modeling, cache filtering, and conservative pseudo-label use [Zhou et al., 2025, Nguyen et al., 2025, Zhai et al., 2025]. FOCUS is closest in spirit to this literature, but it asks a narrower question: does object-centric evidence improve the write decision for STATA relative to those lighter reliability proxies?

Online VLM TTA beyond confidence gating. Several directly relevant VLM TTA methods in the workspace pursue stronger adaptation objectives rather than better write filters. TDA uses a lightweight dynamic cache for efficient training-free adaptation [Karmanov et al., 2024]; BATCLIP performs bimodal online adaptation for CLIP under corruptions [Maharana et al., 2024]; and CLIPTTA replaces entropy-style updates with a contrastive objective aligned with CLIP pretraining [Lafon et al., 2025]. These methods are important context for our negative result: if lightweight gating is insufficient, stronger update rules or multimodal objectives may be the more promising direction.

Selective prediction and abstention. Outside VLM TTA, selective prediction studies when a model should abstain rather than commit. Confidence-based selective classification and integrated reject-option networks provide a broader conceptual backdrop for update gating [Geifman and El-Yaniv, 2017, 2019]. Our problem differs because the decision is not whether to answer, but whether to modify persistent online state. Still, this literature grounds the idea that ranking examples by update safety is a form of selective decision-making rather than a new adaptation mechanism.

Object-centric and context-aware debiasing. Foreground-background decomposition is already established for CLIP robustness and TTA. SegDebias uses segmentation to mitigate bias in CLIP predictions [Wu and Cai, 2025]; GS-Bias learns spatial corrections during single-image TTA [Huang et al., 2025]; Fair Context Learning targets evidence imbalance in VLM adaptation [Yun et al., 2026].

Broader evidence that CLIP-style models overuse contextual features comes from work on task bias, spurious features, and robust context-aware recognition [Menon et al., 2022, Wang et al., 2024, Janouskova et al., 2025]. Our claim is narrower than all of these: not that object-centric reasoning is new, but that it may or may not be useful as an online write-safety signal.

3 Method

3.1 Problem Setting

We consider an online stream $\{x_t\}_{t=1}^{T}$ processed by a frozen CLIP backbone with STATA adaptation state $\mathcal{S}_t$ [Zanella et al., 2025]. At time t, the model predicts a pseudo-label $\hat{y}_t$ from the current state, then either updates or skips:

$$\mathcal{S}_{t+1} = \begin{cases} \mathcal{U}(\mathcal{S}_t, x_t, \hat{y}_t), & g_t = 1, \\ \mathcal{S}_t, & g_t = 0, \end{cases} \tag{1}$$

where $\mathcal{U}$ is the unchanged STATA update rule and $g_t \in \{0, 1\}$ is the gate. The entire paper is about designing and evaluating g_t.

3.2 FOCUS Score

For each image x, FOCUS constructs three views using the same frozen CLIP ViT-B/16 backbone:

1. the original image x_{full};
2. a foreground-focused crop x_{fg};
3. a complementary background image x_{bg}.

Let $z_{\text{full}}, z_{\text{fg}}, z_{\text{bg}} \in \mathbb{R}^C$ be class logits over C classes and let $k = \arg\max_c z_{\text{full},c}$. We define

$$a(x) = \mathbf{1}[\arg\max z_{\text{full}} = \arg\max z_{\text{fg}}], \tag{2}$$

$$f(x) = p_{\text{fg}}(k), \tag{3}$$

$$b(x) = \max_c p_{\text{bg}}(c), \tag{4}$$

where p_{fg} and p_{bg} are the softmax distributions of z_{fg} and z_{bg}. The score is

$$s_{\text{focus}}(x) = a(x)\big(f(x) - \lambda b(x) - \tau\big). \tag{5}$$

Validation tuning selected $\lambda = 1.0$ and $\tau = 0.0$.

3.3 Primary Causal Protocol

The main experiments use only *causal*, validation-derived thresholds. For dataset d, seed s, target rate $r \in \{0.3, 0.5, 0.7\}$, and score function m, we compute validation scores $\{m(x_i)\}_{i=1}^{N_{\text{val}}}$ and define

$$\theta_m^{(d,s,r)} = Q_q\Big(\{m(x_i)\}_{i=1}^{N_{\text{val}}}\Big), \tag{6}$$

with $q = 1 - r$ when larger scores are better and $q = r$ for entropy. Online inference then accepts a sample immediately when its score crosses that frozen threshold. This is the primary deployment-faithful protocol.

Because thresholds are fixed before the test stream starts, the realized acceptance rate is allowed to drift under distribution shift. That drift is a feature, not a bug: it reflects what would happen in an actual online system whose calibration was frozen on validation data.

3.4 Offline Exact-Rate Ranking Protocol

We also retain an *offline oracle-style* protocol for ranking analysis. For a full test stream of length T, we sort the stream scores and accept exactly $K = \text{round}(rT)$ samples. This removes rate drift and isolates how well each score ranks candidate updates, but it is not causal because the accept/reject decision for time t depends on future scores. We therefore use this protocol only to contrast with the causal results; we do not interpret it as realistic online adaptation.

3.5 Foreground and Background Views

Foreground views are derived from CLIP attention rollout over the last six vision-attention blocks, followed by thresholding, largest-component cleanup, and a resized crop. Pilot diagnostics show zero empty-mask fallbacks on both datasets, mean rollout time 16.72 ms on WATERBIRDS and 16.43 ms on COUNTERANIMAL, and mean extra view-construction plus extra encoding cost 40.03 ms and 40.90 ms, respectively. We also include an independent-mask sanity check that rebuilds the views using spectral residual saliency.

3.6 Baselines

We compare FOCUS against four baselines:

- zero-shot CLIP;
- plain STATA with unconditional updates;
- entropy and max-softmax gates;
- CER, a ReTA-style consistency score used only as an update gate.

We also test four simple object-centric ablations: agreement-only, foreground-only, background-only, and no-background ($a(x)f(x)$).

4 Experiments

4.1 Setup

Datasets. We use two context-sensitive benchmarks. WATERBIRDS is loaded from the Hugging Face mirror because no local WILDS installation was available in this workspace. It has two classes and four groups defined by label-place combinations. COUNTERANIMAL is loaded from the Hugging Face mirror and evaluated on deterministic class-balanced validation and test subsets of sizes 1024 and 1536, respectively, because only one split was exposed.

Backbone, streams, and seeds. All runs use `openai/clip-vit-base-patch16` with prompt template `a photo of a {class_name}`. Streams follow the realistic online protocol with $\gamma = 0.3$ and batch size 128. The reporting seeds are $\{7, 17, 37\}$; seed 27 is retained only as a frozen confirmation run and is not averaged into the main tables.

Sparse harmful-update labels. On each validation stream we inspect the first 256 samples, probe every fourth position, and compare the average labeled accuracy over the next 16 samples between an *update* branch and a *skip* branch. An update is labeled harmful if the update branch is worse by at least 0.25 percentage points. This yields 192 direct labels per dataset across the three reporting seeds. Harmful updates are rare: 14/192 positives on WATERBIRDS (7.29%) and 2/192 on COUNTERANIMAL (1.04%).

Metrics. We report online accuracy, balanced accuracy, WATERBIRDS worst-group accuracy, realized acceptance rate, accepted-update precision, direct harmful-update AUPRC/AUROC, and latency. Following reviewer feedback, we distinguish *cached stream latency* from *full end-to-end latency*. For FOCUS, end-to-end latency adds rollout and extra-view encoding overhead to the cached stream-time update loop.

4.2 Causal Threshold Results

Table 1 is the main result. It uses only validation-derived thresholds, so every update decision is causal.

The causal findings are mixed. On WATERBIRDS, FOCUS is the strongest gate by downstream performance: at the nominal 50% rate it reaches 71.03% overall accuracy and 46.24% worst-group accuracy, well above entropy (62.04%, 31.07%), max-probability (62.04%, 31.07%), and CER

Table 1: Primary causal-threshold results at the nominal 50% target rate. All thresholds are calibrated on validation streams and then frozen; realized acceptance rates on test are therefore allowed to drift. Harmful-update AUPRC is computed from the sparse counterfactual labels. End-to-end latency reports the full per-image cost; for FOCUS it adds rollout and extra-view overhead (16.72 ms + 40.03 ms on WATERBIRDS, 16.43 ms + 40.90 ms on COUNTERANIMAL) to the cached stream latency. Best gated result in each dataset block is in **bold**.

Method	WATERBIRDS				COUNTERANIMAL			
	Acc. ↑	Worst-group ↑	Harmful AUPRC ↑	End-to-end ms ↓	Acc. ↑	Bal. Acc. ↑	Realized rate	End-to-end ms ↓
Zero-shot CLIP	**81.93 ± 5.52**	16.43 ± 11.77	0.0169 ± 0.0037	**0.02 ± 0.00**	67.32 ± 0.00	66.41 ± 0.00	0.00	**0.01 ± 0.00**
Plain STATA	72.49 ± 6.17	**48.75 ± 8.26**	0.0169 ± 0.0037	0.32 ± 0.01	62.41 ± 0.59	61.03 ± 0.55	1.00	0.48 ± 0.01
Entropy gate	62.04 ± 6.23	31.07 ± 2.82	0.0285 ± 0.0082	0.29 ± 0.01	**69.44 ± 2.69**	**69.05 ± 2.66**	0.36	0.34 ± 0.00
MaxProb gate	62.04 ± 6.23	31.07 ± 2.82	0.1026 ± 0.0846	0.28 ± 0.01	67.93 ± 1.22	68.00 ± 1.02	0.37	0.35 ± 0.00
CER gate	63.28 ± 7.07	31.66 ± 2.64	**0.1935 ± 0.1349**	0.29 ± 0.00	66.75 ± 2.64	66.45 ± 2.85	0.37	0.35 ± 0.00
FOCUS	**71.03 ± 6.11**	**46.24 ± 7.68**	0.0227 ± 0.0034	57.03	62.30 ± 1.42	61.29 ± 1.44	0.53	57.71

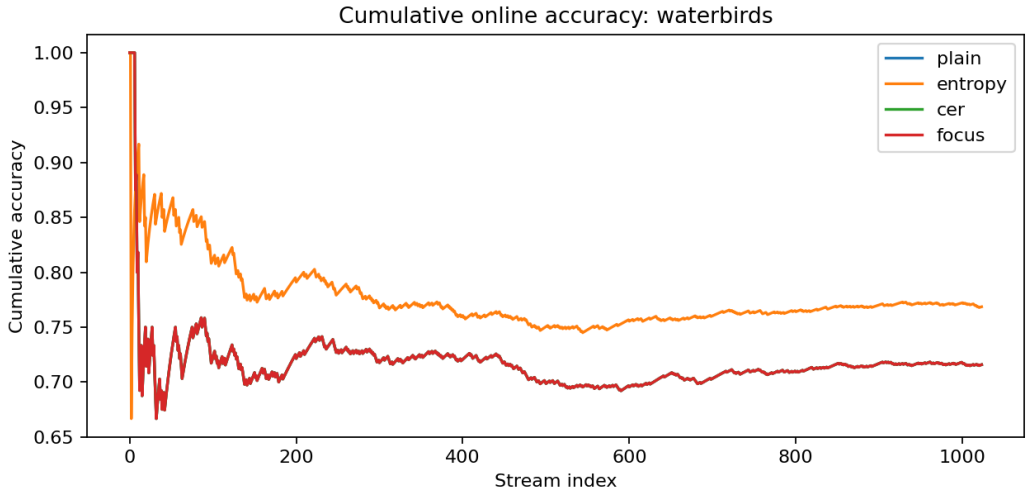

Figure 1: Cumulative online accuracy on the WATERBIRDS test stream at the nominal 50% rate for the median FOCUS reporting seed under the causal threshold protocol. The comparison uses the causal entropy, CER, and FOCUS gates; no offline exact-rate curves are shown here. FOCUS stays closer to plain STATA than entropy or CER on this dataset, but the same pattern does not hold on COUNTERANIMAL.

(63.28%, 31.66%). However, it still trails plain STATA on worst-group accuracy (48.75%) and trails zero-shot CLIP on overall accuracy (81.93%).

On COUNTERANIMAL, the conclusion reverses. FOCUS reaches 62.30% overall accuracy and 61.29% balanced accuracy, below entropy (69.44%, 69.05%), max-probability (67.93%, 68.00%), and CER (66.75%, 66.45%). The thresholded rates also drift differently across methods: entropy, max-probability, and CER all realize only 35.9–37.4% acceptance at the nominal 50% setting, while FOCUS realizes 52.8%. This is exactly why the exact-rate analysis must be treated separately: causal thresholding and offline rate matching answer different questions.

Figure 1 shows the cumulative WATERBIRDS trajectory for the median FOCUS seed under the same causal threshold protocol as Table 1. The qualitative pattern mirrors the table: FOCUS tracks plain STATA more closely than the other causal gates on WATERBIRDS, but this gain does not transfer to COUNTERANIMAL.

4.3 Offline Exact-Rate Ranking Analysis

The exact-rate protocol uses future information and is therefore not a realistic online method. We keep it only as an oracle-style ranking analysis. Appendix A contrasts the causal and exact-rate results at the common nominal 50% rate.

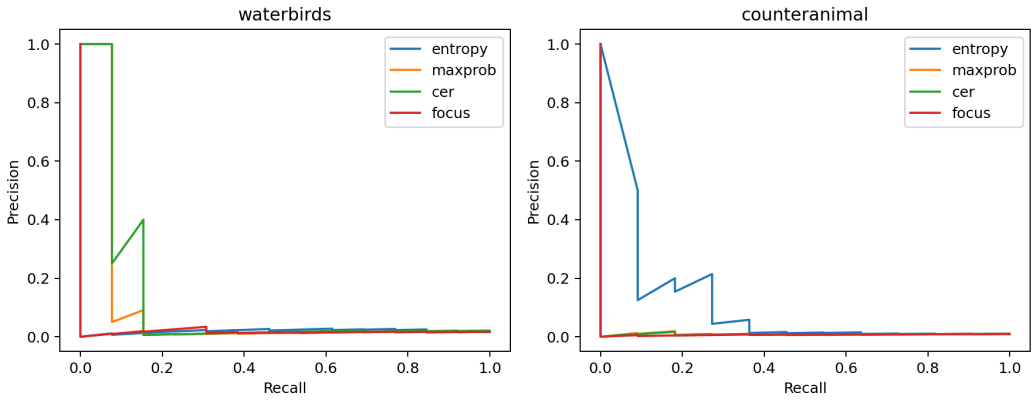

Figure 2: Precision-recall curves for harmful-update ranking on the sparse counterfactual labels. Each curve ranks candidate updates by the raw score before thresholding. On WATERBIRDS, FOCUS improves over entropy but remains below max-probability and CER. On COUNTERANIMAL, the direct label is too sparse for strong conclusions.

The comparison is instructive. On WATERBIRDS, causal and exact-rate results are nearly identical for all methods, so the original conclusion was not driven by the protocol there. On COUNTERANIMAL, exact-rate control removes threshold drift and lifts entropy from 35.9% realized acceptance to exactly 50%, reducing its accuracy from 69.44% to 64.13%. FOCUS changes much less (62.30% causal vs. 62.20% exact), so the exact-rate analysis preserves the same qualitative conclusion: the object-centric gate is not competitive on COUNTERANIMAL even when rate drift is removed. We therefore interpret the exact-rate results as a useful ranking diagnostic, not as deployable online adaptation.

4.4 Does FOCUS Rank Harmful Updates Better?

The answer is no. On the direct harmful-update labels, FOCUS improves over entropy on WATERBIRDS but not over the stronger baselines. In the pilot study, FOCUS reaches direct harmful-update AUPRC 0.142 and AUROC 0.600 on WATERBIRDS, better than entropy (0.062, 0.319) but below max-probability (0.207, 0.681) and CER (0.196, 0.658). In the three-seed main study, WATERBIRDS direct harmful-update AUPRC is only 0.0227 for FOCUS, versus 0.1026 for max-probability and 0.1935 for CER. On COUNTERANIMAL, only 2 of 192 candidate updates are harmful, so all ranking results should be treated cautiously; even there, FOCUS does not emerge as the strongest signal.

4.5 Ablations Under Offline Exact-Rate Control

Table 2 uses offline exact-rate comparisons for the simple variants and hard FOCUS, while the soft-gate row reports its realized near-50% acceptance from the dedicated ablation run. This mixed provenance is explicit because the hard FOCUS exact-rate row lives in the main-study artifact, whereas the soft and independent-mask controls live in the ablation artifact. Under this oracle-style comparison, the full object-centric score is not necessary to reproduce the strongest downstream behavior. The background-only gate is the best simple variant on both datasets. It reaches 79.79% accuracy on WATERBIRDS and 65.84% on COUNTERANIMAL, exceeding hard FOCUS by 8.60 and 3.64 points, respectively. Agreement-only is also stronger than FOCUS on WATERBIRDS.

The soft gate does not materially change the conclusion: on WATERBIRDS it reaches 72.47% accuracy, essentially matching hard FOCUS, while harmful-update AUPRC remains 0.0227. The independent-mask sanity check is also unfavorable. Replacing CLIP rollout with spectral saliency lowers FOCUS to 57.29% on WATERBIRDS and 54.17% on COUNTERANIMAL, indicating substantial sensitivity to the mask source.

Table 2: Ablations at the common nominal 50% setting. Agreement-only, foreground-only, background-only, no-background, independent-mask, and hard FOCUS use offline exact-rate selection, so those rows are ranking-oracle comparisons. The FOCUS soft row instead reports the realized acceptance from its dedicated soft-gating ablation run (0.4977 $\pm$ 0.0158 on WATERBIRDS). Best simple variant in each dataset is in **bold**. The background-only control is the strongest simple ablation downstream on both datasets.

Method	WATERBIRDS Acc. ↑	WATERBIRDS Bal. Acc. ↑	WATERBIRDS Harmful AUPRC ↑	COUNTERANIMAL Acc. ↑	COUNTERANIMAL Bal. Acc. ↑	COUNTERANIMAL Harmful AUPRC ↑
Agreement-only	76.82 ± 1.29	65.19 ± 10.19	0.0175 ± 0.0033	64.39 ± 1.43	63.11 ± 1.57	0.0091 ± 0.0075
Foreground-only	62.60 ± 6.71	66.96 ± 10.84	0.0721 ± 0.0767	62.17 ± 1.71	61.32 ± 1.83	0.0115 ± 0.0102
Background-only	**79.79 ± 3.84**	**69.19 ± 13.79**	**0.0432 ± 0.0199**	**65.84 ± 1.34**	**64.27 ± 1.30**	**0.0269 ± 0.0255**
No-background	62.60 ± 6.71	66.96 ± 10.84	0.0721 ± 0.0767	62.17 ± 1.71	61.32 ± 1.83	0.0117 ± 0.0102
FOCUS soft	72.47 ± 5.54	71.20 ± 9.76	0.0227 ± 0.0034	–	–	–
FOCUS independent mask	57.29 ± 0.74	66.19 ± 0.97	–	54.17 ± 1.47	67.41 ± 1.05	–
Hard FOCUS	71.19 ± 5.98	70.97 ± 10.07	0.0227 ± 0.0034	62.20 ± 1.51	61.10 ± 1.47	0.0089 ± 0.0064

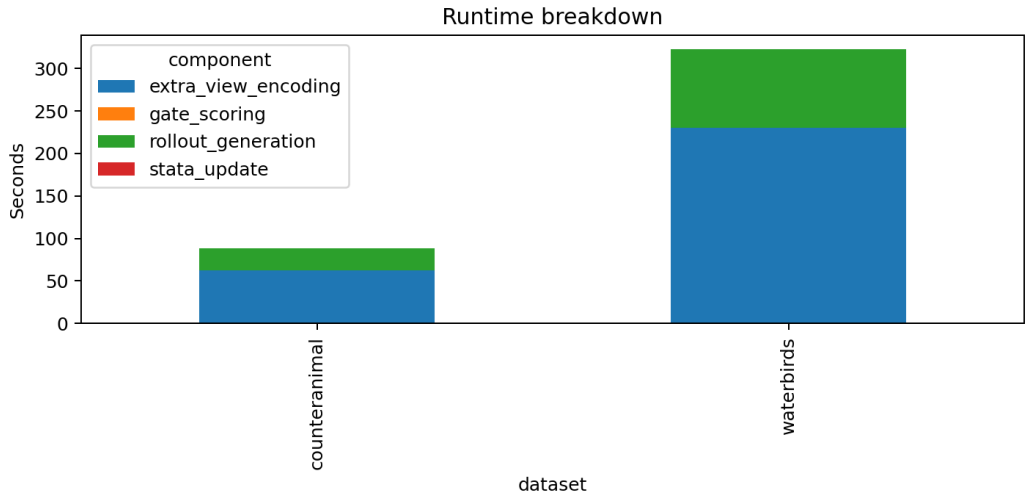

Figure 3: Runtime breakdown for FOCUS on the full seed-7 test stream of each dataset at the common nominal 50% rate. Bars are total wall-clock seconds accumulated over the stream; they are not per-image timings. The stack separates rollout generation, extra-view encoding, gate scoring, and STATA updates, and the dominant cost is the extra object-centric preprocessing rather than gate scoring itself.

4.6 Efficiency and Qualitative Analysis

The original draft compared only cached stream-time latencies, which understated the real cost of FOCUS. The corrected numbers are simple. In the cached online loop, FOCUS costs 0.28 ms per image on WATERBIRDS and 0.39 ms on COUNTERANIMAL, similar to the other gates. But the full method also requires rollout generation and extra foreground/background encoding, adding 16.72 ms + 40.03 ms on WATERBIRDS and 16.43 ms + 40.90 ms on COUNTERANIMAL. The full end-to-end per-image cost is therefore about 57.03 ms on WATERBIRDS and 57.71 ms on COUNTERANIMAL, over two orders of magnitude larger than the cached gate-only number.

Figure 3 visualizes this corrected accounting for one full seed-7 test stream per dataset. The stacked bars report total seconds over the entire stream, not milliseconds per image; dividing those totals by the stream lengths recovers the reported 57.03 ms and 57.71 ms end-to-end per-image costs. The workflow runtime summary is recorded in `exp/report/results.json`: the complete audited workflow, including calibration, main study, ablations, and figure generation, finishes in 6.44 minutes on one RTX A6000 GPU, and the main-and-ablation stage accounts for 5.00 of those minutes. Figure 4 shows representative decompositions. The views are often semantically plausible, but plausibility does not translate into a stronger update-safety ranking.

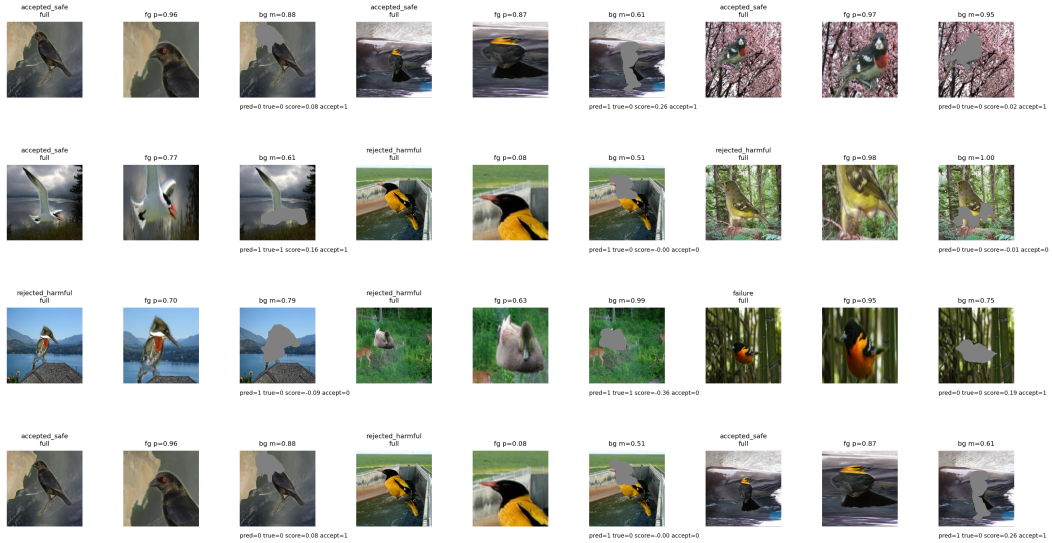

Figure 4: Representative FOCUS decompositions on WATERBIRDS. Each example shows the original image, foreground crop, and complementary background. Although many masks look reasonable, the aggregate experiments show that this evidence is not a reliable standalone write-safety signal.

5 Discussion and Limitations

The corrected paper supports a narrower and more defensible claim than the original draft. Object-centric evidence can help on WATERBIRDS, a dataset where context bias is unusually salient and the rollout masks are derived from the same backbone used for recognition. But that evidence does not yield the best harmful-update ranking, it does not transfer to COUNTERANIMAL, and it is partly reproduced by a much simpler background-only rule.

Several limitations matter for interpretation. First, the benchmark is intentionally small: only WATERBIRDS and COUNTERANIMAL were run under the audited protocol. Second, the direct harmful-update labels are sparse by design, especially on COUNTERANIMAL, where only two positive events are observed. Third, FOCUS uses CLIP-derived masks, and the independent-mask sanity check shows that its behavior is sensitive to that choice. Fourth, SegDebias was not executed in this workspace, so we do not claim a like-for-like comparison with segmentation-heavy debiasing methods. Finally, the causal threshold protocol itself reveals a practical challenge: threshold transfer under shift can drift acceptance rates substantially, which complicates any attempt to deploy a fixed gate across domains.

These limitations do not nullify the paper; they define its value. The main contribution is methodological. Once the evaluation is made causal and the latency accounting is corrected, the object-centric hypothesis looks weaker than its qualitative intuition suggests. That negative result is useful because it narrows where future work should look: stronger uncertainty estimation, better causal diagnostics of update harm, and less circular ways of constructing object evidence.

6 Conclusion

We studied whether object-centric evidence can serve as an update-safety signal for realistic online VLM adaptation. FOCUS uses full-image, foreground, and background views to decide whether STATA should update on the current sample. Under a causal validation-threshold protocol, the result is mixed and overall negative.

On WATERBIRDS, FOCUS is the strongest causal gate downstream, reaching 71.03% online accuracy and 46.24% worst-group accuracy at the nominal 50% rate. On COUNTERANIMAL, it falls to 62.30% and trails all three non-object-centric gates. The mechanistic evidence is also unfavorable: FOCUS

harmful-update AUPRC remains below max-probability and CER, and a background-only ablation outperforms the full score downstream on both datasets. Corrected efficiency reporting further shows that the true end-to-end cost of FOCUS is about 57 ms per image, not the sub-millisecond cached gate-only number.

The broader implication is that visually sensible object cues are not enough by themselves to solve write safety in realistic online adaptation. Future work should treat object-centric evidence as one ingredient in a larger reliability pipeline rather than as a standalone gating rule.

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A Causal vs. Exact-Rate Comparison

Table 3: Causal-threshold versus exact-rate results at the nominal 50% setting. "Exact" denotes the offline oracle-style top-K protocol and should not be interpreted as causal online adaptation.

Method	WATERBIRDS causal	WATERBIRDS exact	COUNTERANIMAL causal	COUNTERANIMAL exact
Entropy	62.04 / 0.51	62.47 / 0.50	69.44 / 0.36	64.13 / 0.50
CER	63.28 / 0.50	63.38 / 0.50	66.75 / 0.37	64.80 / 0.50
FOCUS	71.03 / 0.50	71.19 / 0.50	62.30 / 0.53	62.20 / 0.50

Each cell reports overall online accuracy and realized acceptance rate. The WATERBIRDS numbers change little between protocols, while the COUNTERANIMAL entropy and CER results change substantially because causal threshold transfer drifts their realized rates downward. This is precisely why the exact-rate experiments are best viewed as offline ranking analysis rather than deployable online adaptation.

B Reproducibility Summary

Datasets and streams. WATERBIRDS is evaluated from the Hugging Face mirror with its standard two-class, four-group label-place structure. COUNTERANIMAL is evaluated from the Hugging Face mirror using deterministic class-balanced validation and test subsets of sizes 1024 and 1536. All streams use the realistic online construction with $\gamma = 0.3$ and batch size 128.

Seeds. The reporting seeds are $\{7, 17, 37\}$. Seed 27 is reserved for confirmation runs only and is not averaged into the main tables. Harmful-update pilots use the same reporting seeds.

Threshold selection. The primary causal protocol uses frozen validation thresholds at target rates $\{0.3, 0.5, 0.7\}$. For FOCUS, validation selects $\lambda = 1.0$, $\tau = 0.0$, and the common nominal comparison rate 0.5 using mean validation direct harmful-update AUPRC. Exact-rate results instead sort the test-stream scores offline and accept exactly $K = \mathrm{round}(rT)$ samples.

Result files backing the manuscript. Table 1 uses `exp/causal_threshold/results.json`. Table 2 has mixed row provenance, recorded explicitly in `exp/report/results.json`: agreement-only, foreground-only, background-only, no-background, FOCUS soft, and FOCUS independent-mask come from `exp/ablations/results.json`, while hard FOCUS comes from the exact-rate `focus` row in `exp/main_study/results.json`. Figure 2 and Figure 4 are generated during the report stage recorded in `exp/report/results.json`. Figure 1 uses the causal WATERBIRDS cumulative plot in `figures/waterbirds_cumulative_accuracy.png`. Figure 3 uses `figures/runtime_breakdown.png`. The workflow runtime summary cited in the paper is stored in `exp/report/results.json` and points to the stage-level accounting in `artifacts/results/runtime_accounting.json`. The consolidated audit artifact `results.json` mirrors the headline rows from the stage-specific files and records the overall claim check.